Nixie Calculator User Manual

(RPN Version)

Nixie Calculator User Manual (RPN)

Contents

General information

Apart from the power supply, the nixie calculator works completely independently and requires neither a network connection nor any peripheral devices.
This manual does not describe how to use an RPN calculator. There are numerous tutorials available online. Just this: to calculate 1 + 2 type [1] [ENT] [2] [+]

Power supply

The calculator needs a DC 12V/2A **regulated** power supply with a center positive barrel jack plug (5.5/2.1mm compatible).

Power on / Power off

Use the power switch to the left of the keyboard to power on and off the calculator. During start-up, the calculator briefly shows the controller firmware version on the left and the keyboard firmware version on the right. By default, the calculator starts in calculator mode.

The [F] key

The [F] key has several functions:

- Press and release the [F] key to switch between the calculator and the clock mode and to leave the menu mode.
- Hold the [F] key for 3 seconds to enter the menu mode.
- Press the [F] key + an operator key to access the lower functions in calculator mode, for example x^2
- Press the [F] key + some defined keys to get a shortcut for some settings, for example LED lighting (see Table of shortcuts).

The [↑] key

Press the [↑] key + an operator key to access the upper functions in calculator mode, for example n!

Calculator mode

In this mode the device works like an RPN calculator (see List of operations). However, be aware that the arbitrary-precision arithmetic used by the calculator is experimental.
The precision of the calculations and of the internal registers has been set to 32 as a compromise between accuracy and performance. Some calculations with very big numbers can be slow. Exponents can be entered and displayed between -9999 and 9999.
If an error occurs (e.g. overflow, domain, divide by zero), an error code is shown in the center of the display (see List of error codes).

Clock mode

Entering date and time

In clock modes (0-8) press the [CLS] key. A blinking zero indicates that you can enter the date and the time in the YYYYMMDDhhmmss format. The [←] key deletes the last entered digit. Press [ENT] to confirm or [CLS] to abort.

Clock modes
There are several clock modes. They can be reached directly with the [0] to [9] keys and the [00] key for the stopwatch mode:

Clock modes

0 - time only
1 - time with no seconds
2 - moving time
3 - time or date
4 - time and date
5 - time and temperature
6 - time and date and temperature
7 - raw date and time
8 - dual time

Special clock modes

9 - timer
10 - stopwatch

Timer
In timer mode press the [CLS] key. A blinking zero indicates that you can enter the number of days, hours, minutes, and seconds in the format DDhhmmss. The [←] key deletes the last entered digit. Press [ENT] to confirm or [CLS] to abort. Press [ENT] to start and stop the timer. Press [←] to reset the timer. Blinking LEDs indicate the end of the countdown; there is no sound. The flashing can be stopped by pressing the [CLS] key. The accuracy depends on the internal MCU oscillator.

Stopwatch
In stopwatch mode press the [ENT] key to start the stopwatch. Press [ENT] to pause the display while the stopwatch keeps running. Press [←] to reset the stopwatch. The accuracy depends on the internal MCU oscillator.

Menu mode
Hold the [F] key for 3 seconds to enter the menu mode. The setting ID (see Settings table) is displayed on the left, the setting value(s) on the right. Key autorepeat is enabled in menu mode and starts after 1 second. The autorepeat speed increases after some time. Press the [F] key to leave the menu mode and store the values.

Navigation

Keys	Description
[STO]	Next setting
[RCL]	Previous setting
[+]	Next value
[-]	Previous value
[ENT]	Accept value and move to the next column if any
[←]	Restore to previously stored value
[CLS]	Reset to default value
[F] + [CLR]	Reset all settings to default
[F] + [CLS]	Exit menu mode and discard changes
[F]	Exit menu mode and commit changes

Settings table

ID	Name	Description	Values
1	startupmode	Start in calculator or in clock mode	0 = calculator (default) 1 = clock
2	showversion	Show version at startup	0 = off 1 = on (default)
3	autooffmode	Auto off action after a period of no keyboard activity	0 = auto off disabled 1 = shutdown high voltage 2 = switch to clock mode (default)
4	autooffdelay	Delay in minutes for auto off mode	1 - 720 (default 5)
5	clockmode	Initial clock mode	0 = time (default) 1 = time, no seconds 2 = moving time 3 = time or date 4 = time and date 5 = time and temperature 6 = time and date and temperature 7 = raw date and time 8 = dual time 9 = timer 10 = stopwatch
6	hourmode	12 or 24 hours mode	0 = 12 hours 1 = 24 hours (default)
7	leadingzero	Show hours leading zero	0 = off 1 = on (default)
8	timeseparator	Separator mode in compact time format	0 = off 1 = blink (default) 2 = on
9	dateformat	Date format	0 = ddmmyy (default) 1 = yymmdd 2 = mmddyy 3 = yyddmm
10	pirmode	Use PIR to reduce the operating time of the nixie tubes	0 = off (default) 1 = on
11	pirdelay	PIR delay time in minutes before shutting down the high voltage	1 - 720 (default 5)
12	gpsmode	Sync with GPS time	0 = off (default) 1 = on
13	gpsspeed	GPS communication baud rate	0 = 2400 1 = 4800 2 = 9600 3 = 19200 4 = 38400 (default) 5 = 57600 6 = 115200
14	gpssyncinterval	GPS time sync interval in minutes	1 - 720 (default 10)
15	temperaturemode	Use temperature sensor	0 = off (default) 1 = on
16	temperaturecf	Temperature in C or F	0 = Celsius (default) 1 = Fahrenheit
17	ledmode	LEDs on by time or always *	0 = time 1 = always (default)
18	calcrgbmode	RGB mode in calculator mode *	0 = off (default) 1 = by content 2 = by content (all digits) 3 = random (turned on digits only) 4 = full random (turned on digits only) 5 = fixed color (all digits) 6 = random (all digits) 7 = full random (all digits)

19	clockrgbmode	RGB mode in clock mode *	0 = off (default) 1 = by content 2 = random (turned on digits only) 3 = full random (turned on digits only) 4 = fixed color (all digits) 5 = random (all digits) 6 = full random (all digits)
20	trigcolorchange	Trigger color change for random RGB modes in clock mode * (this setting does not apply to the “moving time” clock mode as the color changes with every movement)	0 = off (default) 1 = every second 2 = every minute 3 = every hour
21	ledstarttime	Start time of LED lighting *	00:00 - 23:59 (default 00:00)
22	ledduration	Duration in minutes of LED lighting *	0 - 720 (default 0)
23	ledstarttime2	Start time of LED lighting *	00:00 - 23:59 (default 00:00)
24	ledduration2	Duration in minutes of LED lighting *	0 - 720 (default 0)
25	acpstarttime	Start time of cathode poisoning prevention	00:00 - 23:59 (default 00:00)
26	acpduration	Duration in minutes of cathode poisoning prevention	0 - 720 (default 0)
27	acpforceon	Force turning nixies on during cathode poisoning prevention	0 = off 1 = on (default)
28	negativecolor	RGB LED color for negative numbers in calculator mode *	0-255,0-255,0-255 (default 0,0,0)
29	positivecolor	RGB LED color for positive numbers in calculator mode *	0-255,0-255,0-255 (default 0,0,0)
30	errorcolor	RGB LED color for error in calculator mode *	0-255,0-255,0-255 (default 0,0,0)
31	negexpcolor	RGB LED color for negative exponents in calculator mode *	0-255,0-255,0-255 (default 0,0,0)
32	posexpcolor	RGB LED color for positive exponents in calculator mode *	0-255,0-255,0-255 (default 0,0,0)
33	fixedcalccolor	RGB LED fixed color in calculator mode *	0-255,0-255,0-255 (default 0,0,0)
34	timecolor	RGB LED color for time in clock mode *	0-255,0-255,0-255 (default 0,0,0)
35	timecolor2	RGB LED color for dual time in clock mode *	0-255,0-255,0-255 (default 0,0,0)
36	datecolor	RGB LED color for date in clock mode *	0-255,0-255,0-255 (default 0,0,0)
37	tempcolor	RGB LED color for temperature in clock mode *	0-255,0-255,0-255 (default 0,0,0)
38	fixedcolor	RGB LED fixed color in clock mode *	0-255,0-255,0-255 (default 0,0,0)
39	dstweek	Daylight saving time change, week of month	1 = first 2 = second 3 = third 4 = fourth 5 = last (default)
40	dstdow	Daylight saving time change, day of week	0 = sunday (default) - 6 = saturday
41	dstmonth	Daylight saving time change, month	0 = jan - 11 = dec (default 2 = mar)
42	dsthour	Daylight saving time change, hour	0 - 23 (default 2)
43	dstoffset	Daylight saving time change, offset to UTC in minutes	-720 - 840 (default 120)
44	stdweek	Standard time change, week of month	1 = first 2 = second 3 = third 4 = fourth 5 = last (default)
45	stddow	Standard time change, day of week	0 = sunday (default) - 6 = saturday
46	stdmonth	Standard time change, month	0 = jan - 11 = dec (default 9 = oct)
47	stdhour	Standard time change, hour	0 - 23 (default 3)
48	stdoffset	Standard time change, offset to UTC in minutes	-720 - 840 (default 60)
49	dstweek2	Dual time daylight saving time change, week of month	1 = first 2 = second 3 = third 4 = fourth 5 = last (default)
50	dstdow2	Dual time daylight saving time change, day of week	0 = sunday (default) - 6 = saturday

51	dstmonth2	Dual time daylight saving time change, month	0 = jan - 11 = dec (default 2 = mar)
52	dsthour2	Dual time daylight saving time change, hour	0 - 23 (default 2)
53	dstoffset2	Dual time daylight saving time change, offset to UTC in minutes	-720 - 840 (default 0)
54	stdweek2	Dual time standard time change, week of month	1 = first 2 = second 3 = third 4 = fourth 5 = last (default)
55	stddow2	Dual time standard time change, day of week	0 = sunday (default) - 6 = saturday
56	stdmonth2	Dual time standard time change, month	0 = jan - 11 = dec (default 9 = oct)
57	stdhour2	Dual time standard time change, hour	0 - 23 (default 3)
58	stdoffset2	Dual time standard time change, offset to UTC in minutes	-720 - 840 (default 0)
59	gpsnotifysync	Notify GPS time sync with a short LED flash *	0 = off (default) 1 = on
60	gpssynccolor	RGB LED color for notifying GPS time sync *	0-255,0-255,0-255 (default 255,0,0)
61	notifytimer	Notify end of timer with flashing LEDs *	0 = off 1 = on (default)
62	timercolor	RGB LED color for notifying end of timer *	0-255,0-255,0-255 (default 255,255,255)
63	fixeddecimals	Number of fixed decimals	0 = floating (default) 1 - 8 = number of fixed decimals
64	anglemode	Startup angle mode	0 = degrees (default) 1 = radians
65	showbusycalc	Show animation during long calculations	0 = off 1 = moving decimal point (default) 2 = digit flickering
66	maxexplength	Max length of the exponent	2 - 4 (default 4)
67	scrolldelay	Interval while scrolling result in 1/10 seconds	1 - 20 (default 5)
68	precision	Calculator precision (restart needed)	20 - 32 (default 32)
69	calcinputdirec	Calculator input direction and output format	0 = left to right (default) 1 = right to left 2 = right to left with zero padding
70	inputblinking	Blinking behavior during time, timer and menu value input	0 = off 1 = on (default)
71	brightness	Display brightness: 15 disables PWM dimming on nixie displays	1 - 15 (default 15)
72	dimbrightness	Display brightness during the dimming period	1 - 15 (default 1)
73	dimstarttime	Start time of display dimming	00:00 - 23:59 (default 00:00)
74	dimduration	Duration in minutes of display dimming	0 - 720 (default 0)
75	apautostart	Enable the access point and the web server at startup **	0 = off (default) 1 = on
76	rtcdriftcorr	RTC drift compensation in seconds per month	-60 - 60 (default 0)
77	exttempcorr	External temperature sensor correction in 0.1 °C	-100 - 100 (default 0)

(*) not available for 7-segment LED version
(**) only available if WEB_SUPPORT is set to true

Web server

If enabled in the firmware, press [F] + [ENT] to start an ESP32 Wi-Fi access point (AP) and a web server. After starting, the IP address of the server is displayed for 2 seconds and the network activity LED lights up. By default, the IP address is 192.168.4.1. You can now connect with a smartphone or a PC to this AP and open the site (http://192.168.4.1) with a browser.

- [Calculator] shows a fully functional keypad and mirrors the display. This page is WebSocket based and there is no polling or browser refresh needed.
  There are 4 buttons:
  - [Clock/Calculator/Exit Menu] toggles the device mode or exits the menu mode
  - [Menu] enters the menu mode
  - [Show/Hide Registers] shows/hides all the stack and memory registers with all the digits. The X register holds the current result.
  - [Back] back to homepage

- [Configuration] provides full access to the configuration of the calculator.

- [Status] displays some status information.

- [Time Sync] provides synchronization of the calculator time with the browser time.

- [SSID & Passwords] change the SSID, the AP password and the firmware update password

- [Firmware Update] OTA firmware update

Be aware that the browser connection is not encrypted and the clear text passwords and the SSID of the AP are stored in flash memory and are also visible in the source code. While connected to this AP, you may have no Internet connection. Press [F] + [ENT] again to stop the server and the AP.

Peripherals module

If you have assembled the peripherals module you can connect it to the calculator with a **straight** ethernet patch cable. **Turn off the calculator while connecting or disconnecting the ethernet cable.**
The default communication speed for the BE-220 GPS module is 38400 and 9600 for the older BN-220 module.

Reset SSID and passwords

You can reset the SSID, the AP password and the firmware update password to the initial values (defined in the source code) by pressing [F] + [←] in menu mode. This only works if using the physical keyboard.

“Factory” reset

Press and hold the button on the back for about 4 seconds during the startup. If the reset is successful, all decimal places will flash 5 times. No restart is required after the reset.

OTA firmware update

If enabled in the firmware configuration, the firmware can be updated “over-the-air” using the web interface:

- Open the web interface in your browser as described in the “Web server” section.
- Press [F] + [±] to enable OTA firmware update for a few minutes. During this time, the net activity LED will blink.
- Go to the firmware update page
- Enter the firmware update password

- Select the firmware file
- Select [Upload]

After the update, the calculator restarts automatically.

Troubleshooting

If the calculator loses the time, please change the CR2032 battery.

Table of shortcuts

Keys	Description	Mode
[F] + [←]	Switch LED lighting mode, overrides the time constraints (*)	Calculator, Clock
[F] + [CLS]	Restore the lighting time constraints	Calculator, Clock
[F] + [STO]	Commit temporarily changed settings	Calculator, Clock
[F] + [ENT]	Start/stop Wi-Fi access point and web server	Calculator, Clock
[F] + [±]	Enable/disable OTA firmware update	Calcualtor, Clock
[F] + [0] – [8]	Change the number of fixed decimals, 0 = floating (*)	Calculator
[F] + [9]	Change the input direction and output format (*)	Calculator
[F] + [EXP]	Force scientific notation on or off (*)	Calculator
[F] + [.]	Start/stop scrolling additional result digits	Calculator
[F] + [+]	Increase brightness (*)	Calculator
[F] + [-]	Decrease brightness (*)	Calculator
[F] + [00]	Trim X register to displayed value	Calculator
[0] - [9],[00]	Switch the clock mode (*)	Clock
[F] + [+]	Adjust the time by plus one second	Clock
[F] + [-]	Adjust the time by minus one second	Clock
[F] + [0]	Display free heap memory and minimum free heap memory	Clock
[F] + [00]	Display firmware versions	Clock
[F] + [.]	Display board temperature	Clock
[F] + [RCL]	Display uptime in format ddddd hh mm	Clock
[F] + [CLR]	Reset all settings to default	Menu
[F] + [CLS]	Exit menu mode and discard changes	Menu
[F] + [←]	Reset Wi-Fi access point password	Menu

(*) temporarily, changed settings are not committed

List of error codes

Error Code	Description
1	Overflow
2	Divide by zero
3	Domain error / Invalid input
4	Out of memory
5	Indefinite result
6	Invalid range
7	No result
8	Unknown operation
9	Unknown error

List of operations

Operation	Description
EXP	Enter exponent
clx	Clear X register
←	Clear last entered digit / clear X register if not in input mode
CLS	Clear stack and lastX
↑	Shift key
F	Function key
x^y	Power
x^2	Square
$\sqrt[y]{x}$	Root
x^3	Cube
1/x	Reciprocal
n!	Factorial
e^x	Exponential
ln	Natural logarithm
e	e
mod	Modulo
$\log_y$	Logarithm base y
py,x	Permutations
int	Integer portion
sin	Sine
$\sin^{-1}$	Arcsine
$\sin_h$	Hyperbolic sine
cos	Cosine
$\cos^{-1}$	Arccosine
$\cos_h$	Hyperbolic cosine
tan	Tangent
$\tan^{-1}$	Arctangent
$\tan_h$	Hyperbolic tangent
log	Logarithm base 10
π	Pi
rnd	Pseudorandom number
d↔r	Switch between degrees and radians
cy,x	Combinations
Δ%	Percent difference
±	Change sign
√	Square root
%	Percent
÷	Division
x	Multiplication
-	Subtraction
+	Addition
X↔Y	Swap X and Y registers
lstx	Load lastX into the X register
R↓	Roll down stack
R↑	Roll up stack
CLR	Clear all memory registers
STO	Store into memory register
RCL	Recall from memory register